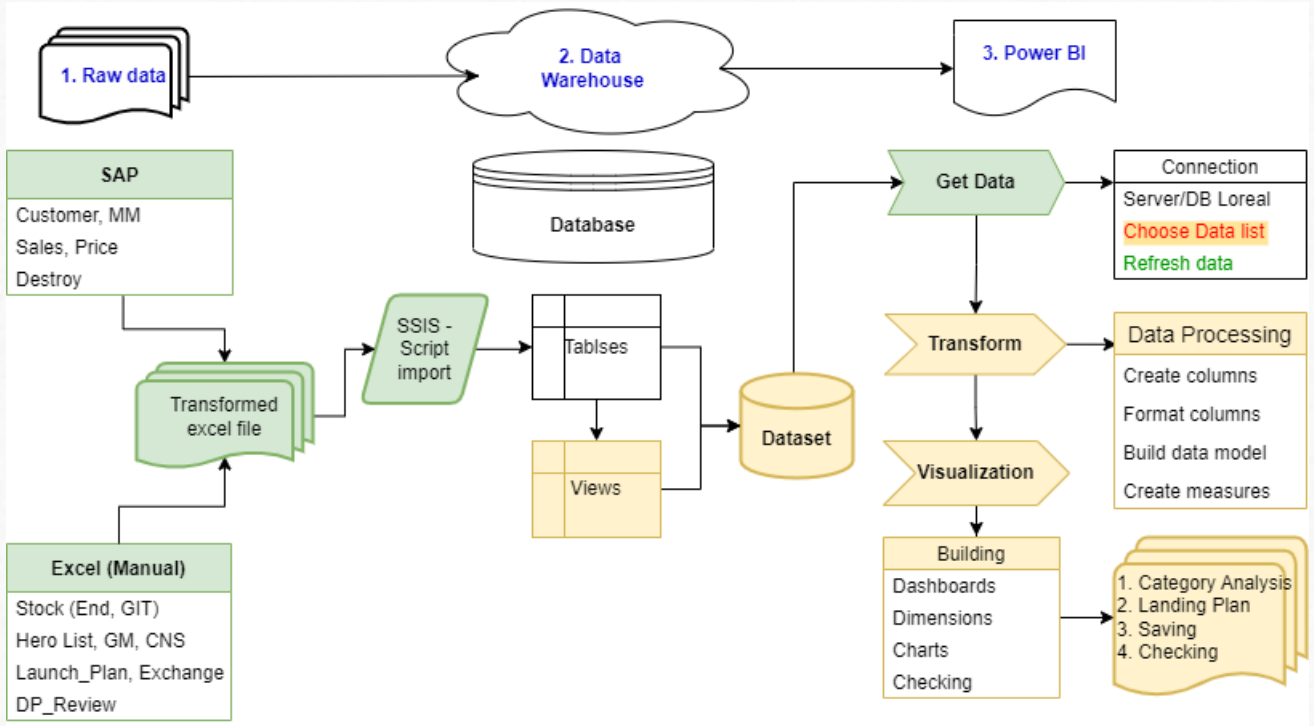
1. Forecast BI Project – Data Flow



The system architecture of the Forecast BI project shows data flow and storage in steps.

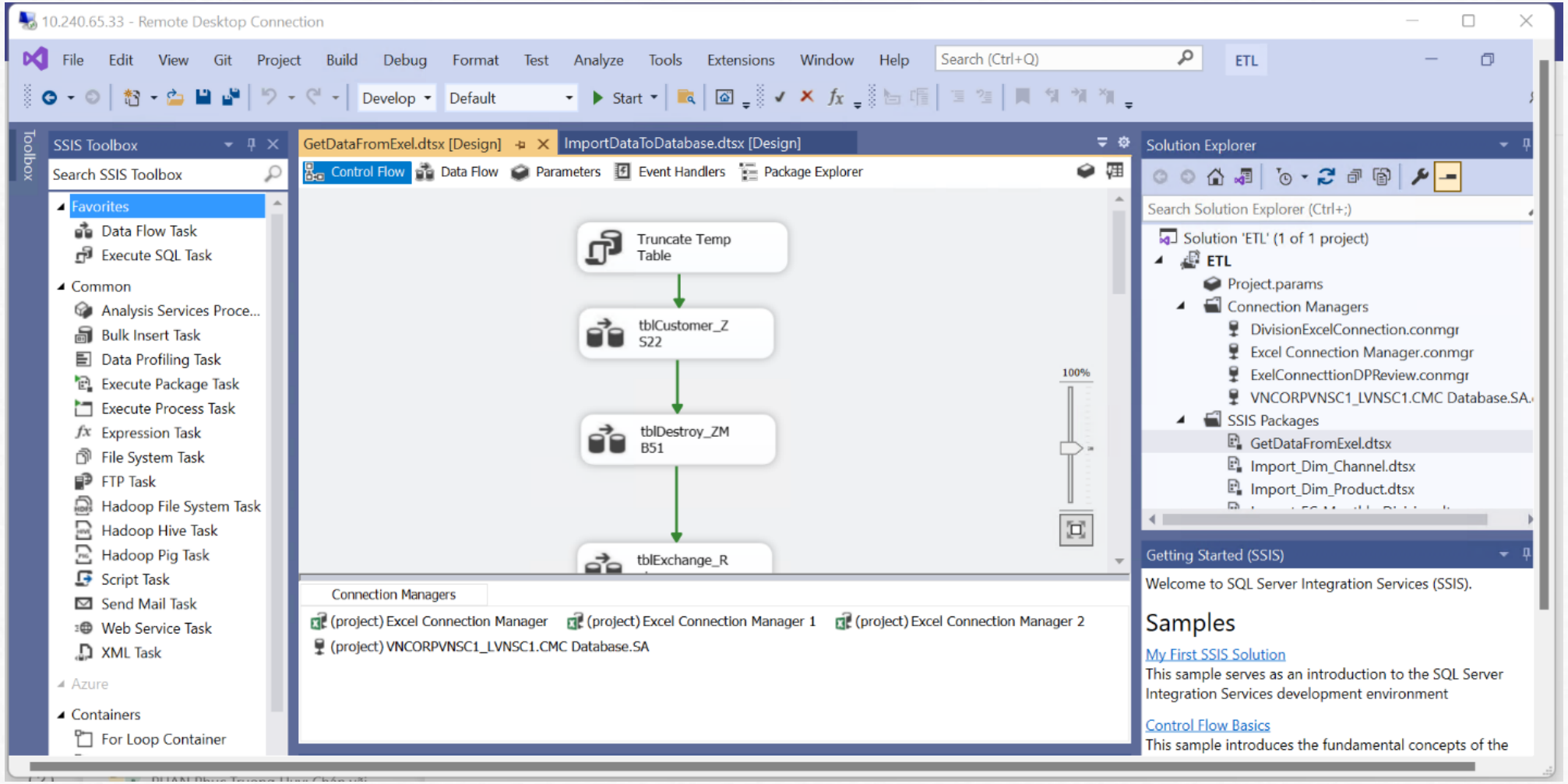
The transformed Excel file gets raw data from the right list and provides meaningful fields for the data warehouse using the SSIS tool.

Views contain data for building BI reports in divisions.

STT	Tên file	Loại file	No.	Folder transform
1	ACD_PRICE_CRV	.TXT	1	tblPrice_YR03
2	ACD_PRICE_LRP	.TXT		
3	ACD_PRICE_SKC	.TXT		
4	ACD_PRICE_VIC	.TXT		
5	CPD_PRICE_ALL	.TXT		
6	LPD_PRICE_ALL	.TXT		
7	PPD_PRICE_C2	.TXT		
8	PPD_PRICE_C4	.TXT		
9	MM_ZMR54OLD	.TXT	2	tblMM_ZMR54
10	MM_ZMR54OLD_Y8	.TXT		
11	ZS22	.TXT	3	tblCustomer_ZS22
12	ZV14_MONTHLY_02	.TXT	4	tblSale_ZV14_02
13	ZV14_MONTHLY_Y8	.TXT	5	tblSale_ZV14_Y8
14	Destroy_ZMB51	.TXT	6	tblDestroy_ZMB51
15	Stock_End	.XLSX	7	tblStock_End
16	Stock_GIT	.XLSX	8	tblStock_GIT
17	Template Hero SKU - ACD	.XLSX	9	tblHero_List
18	Template Hero SKU - CPD	.XLSX		
19	Template Hero SKU - LPD	.XLSX		
20	Template Hero SKU - PPD	.XLSX		
21	tblExchange_Rate	.XLSX	10	tblExchange_Rate
22	tblGross_Margin	.XLSX	11	tblGross_Margin
23	tblDP_Review	.XLSX	12	tblDP_Review
24	tblLaunch_Plan	.XLSX	13	tblLaunch_Plan
25	tblCNS	.XLSX	14	tblCNS

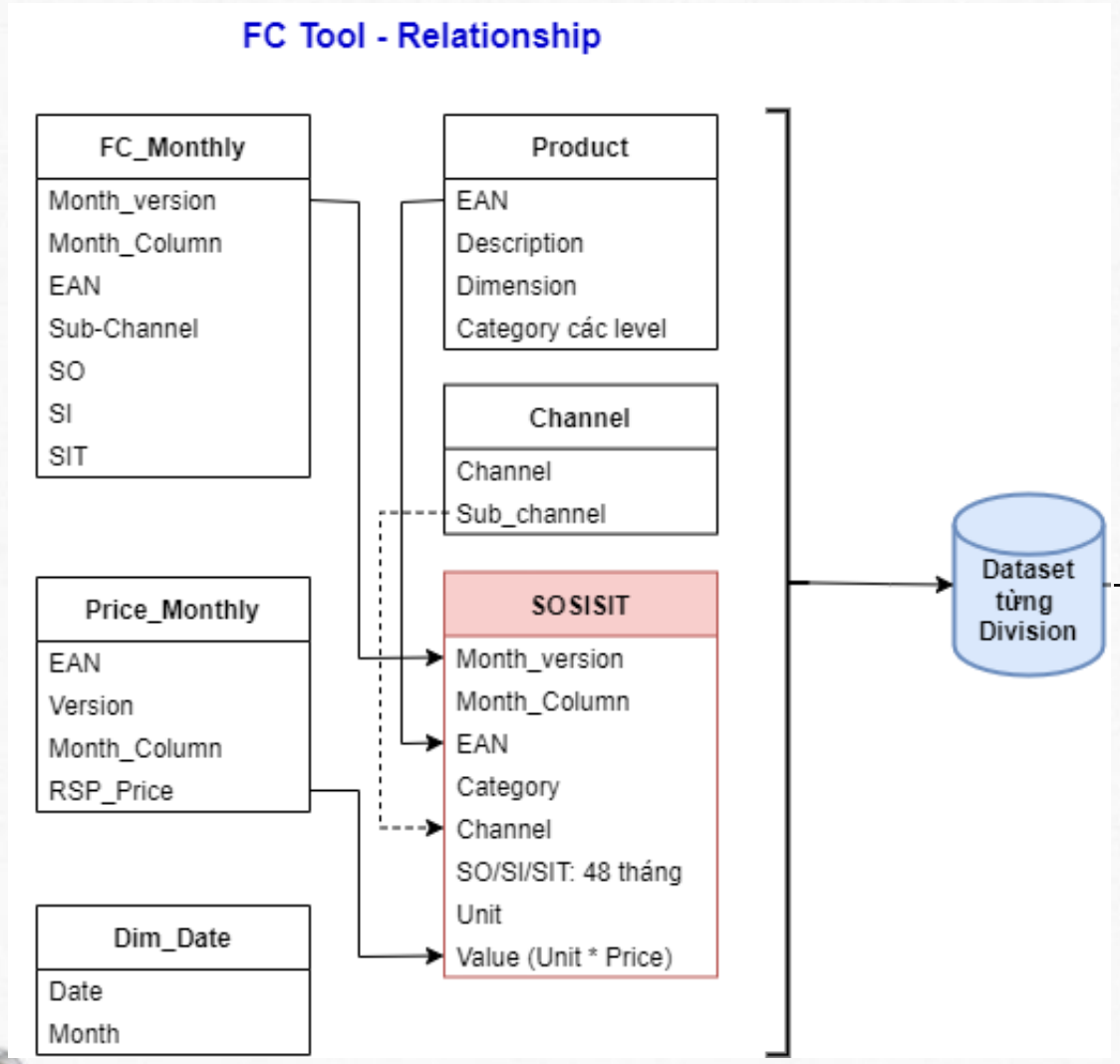
1. Forecast BI Project - ETL

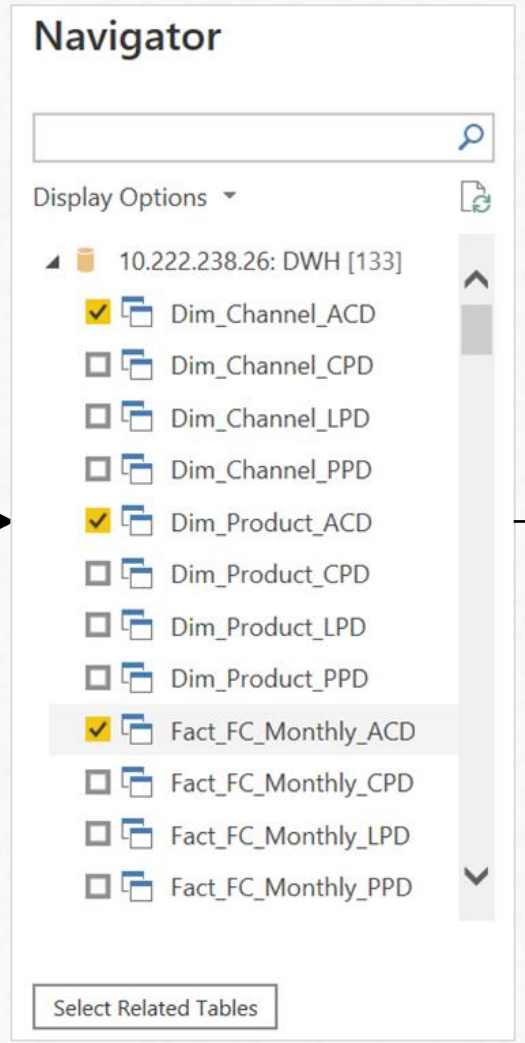
The SSIS tool is chosen for ETL in this project. It requires setting up the data flow task and executing the SQL task for the sequence process in loading and transformation.



1. Forecast BI Project – DWH

The data model is designed in DWH. It provides suitable data for divisions to use in building BI reports.

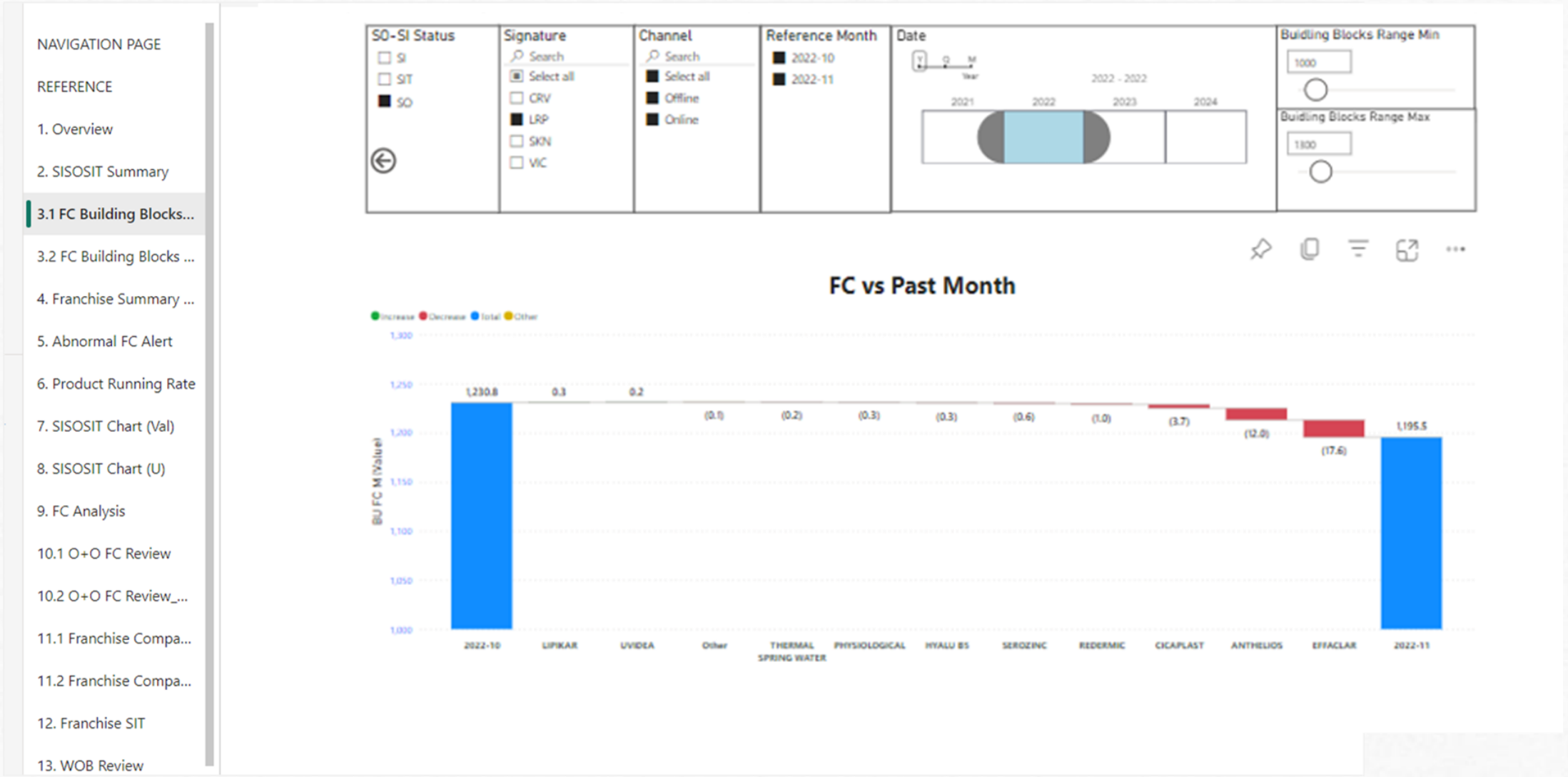






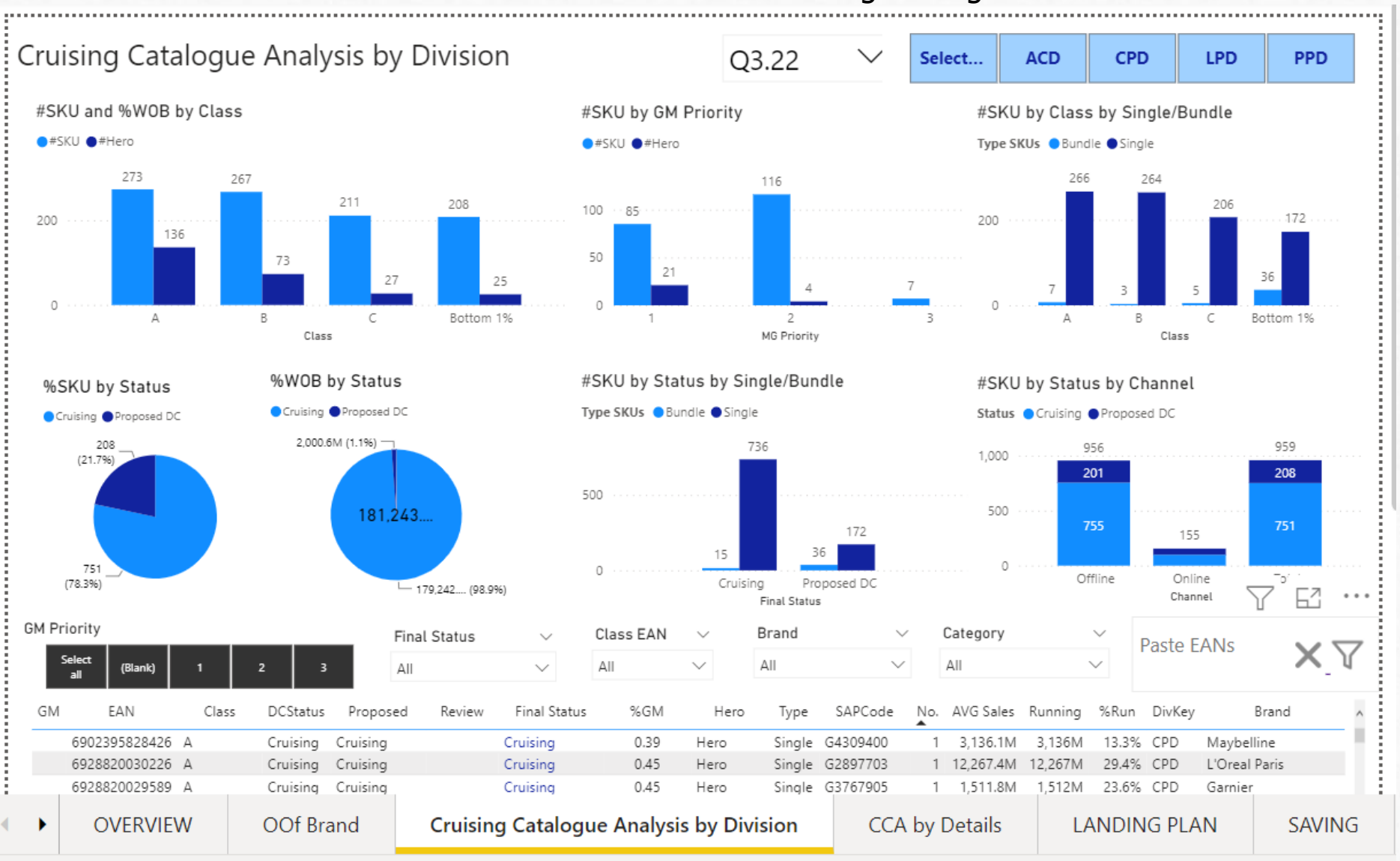
1. Forecast BI Project – Variance Report

The list of pages in the BI report provides dashboards to gain insights into the organization.



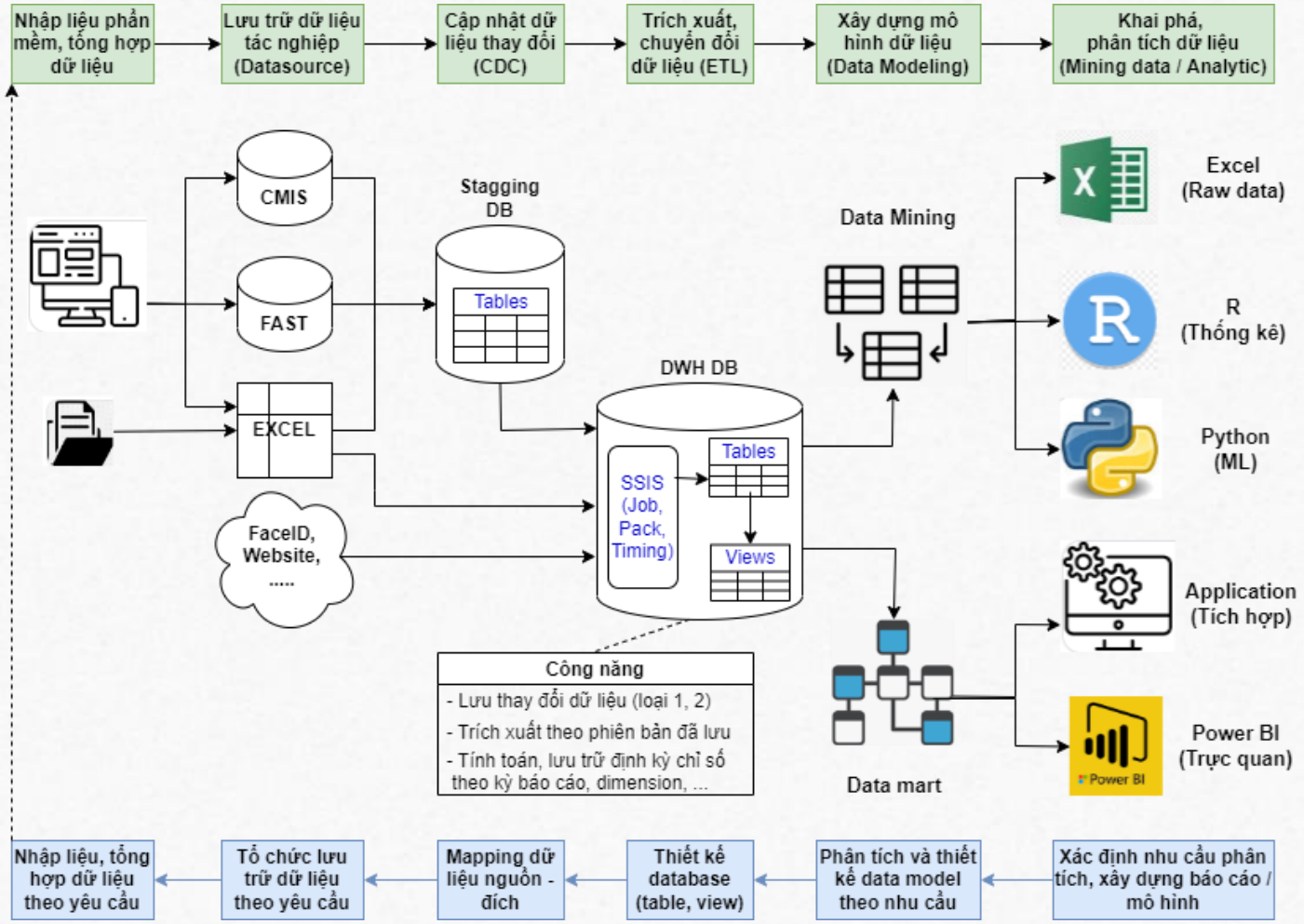
1. Forecast BI Project - Category Management

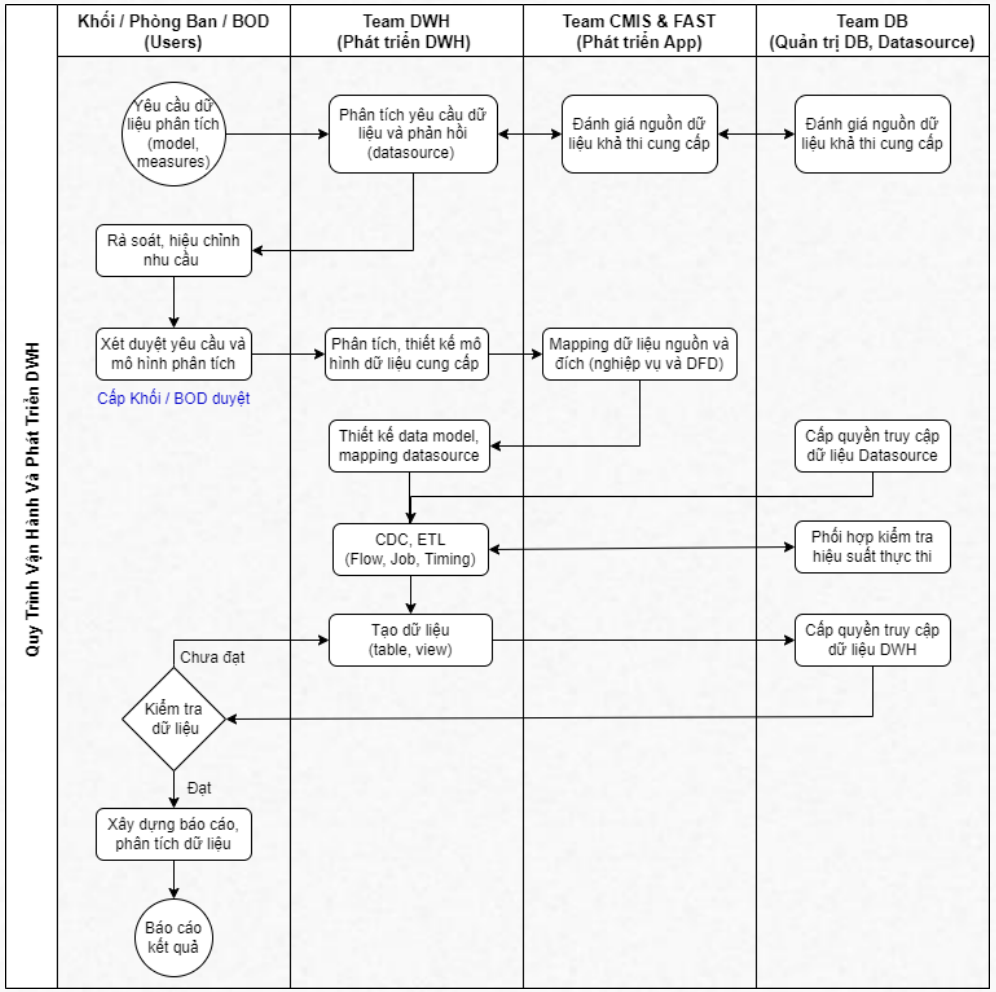
The dashboard shows measures in catering management.



2. Performance BI Project – DWH Operation

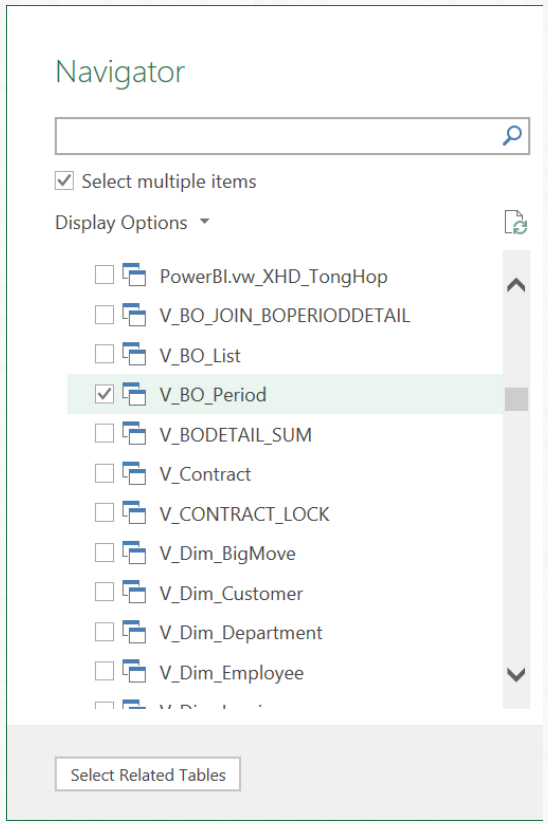
A process of building and maintaining DWH for mining data. Some method for mining data has been determined.

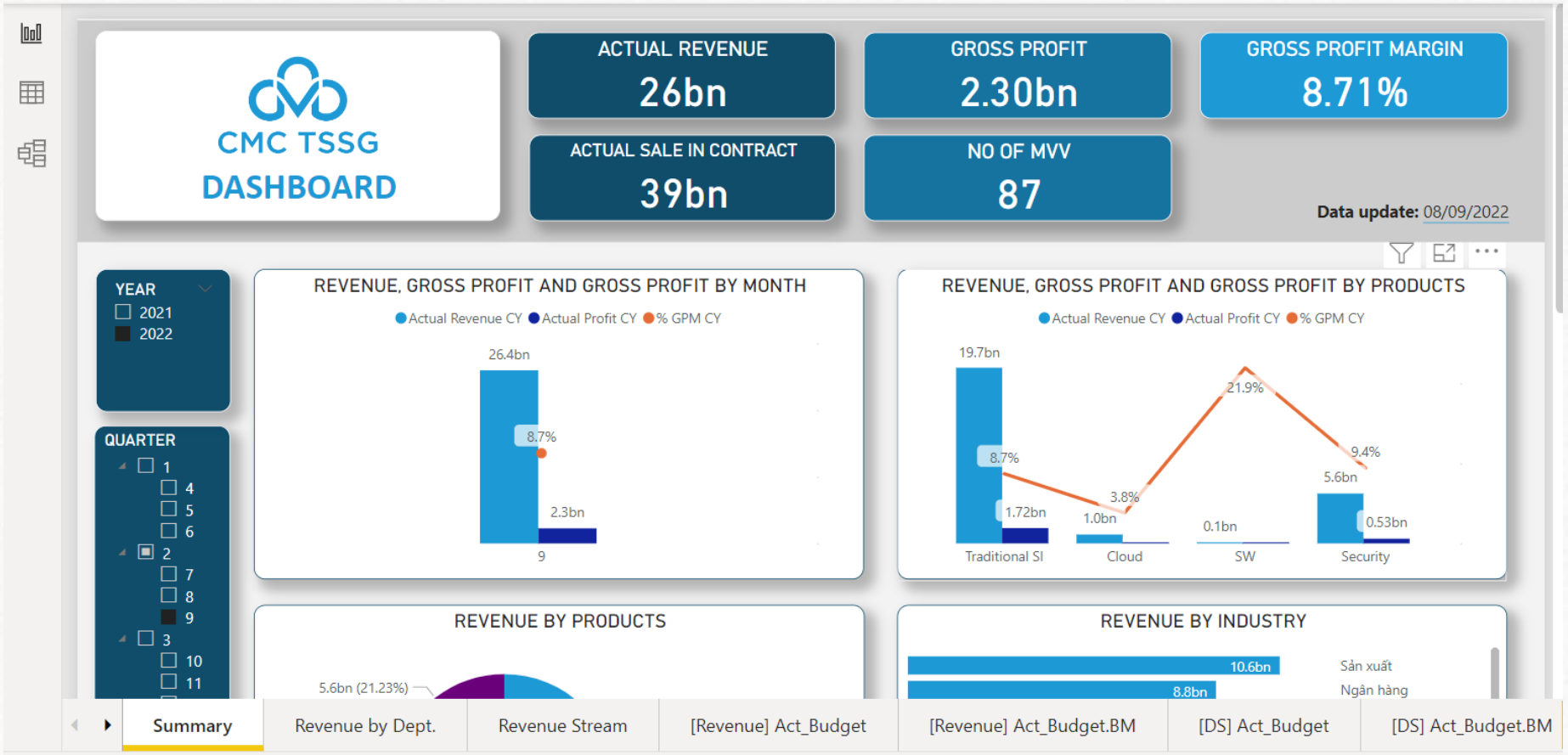




2. Performance BI Project - Dashboard

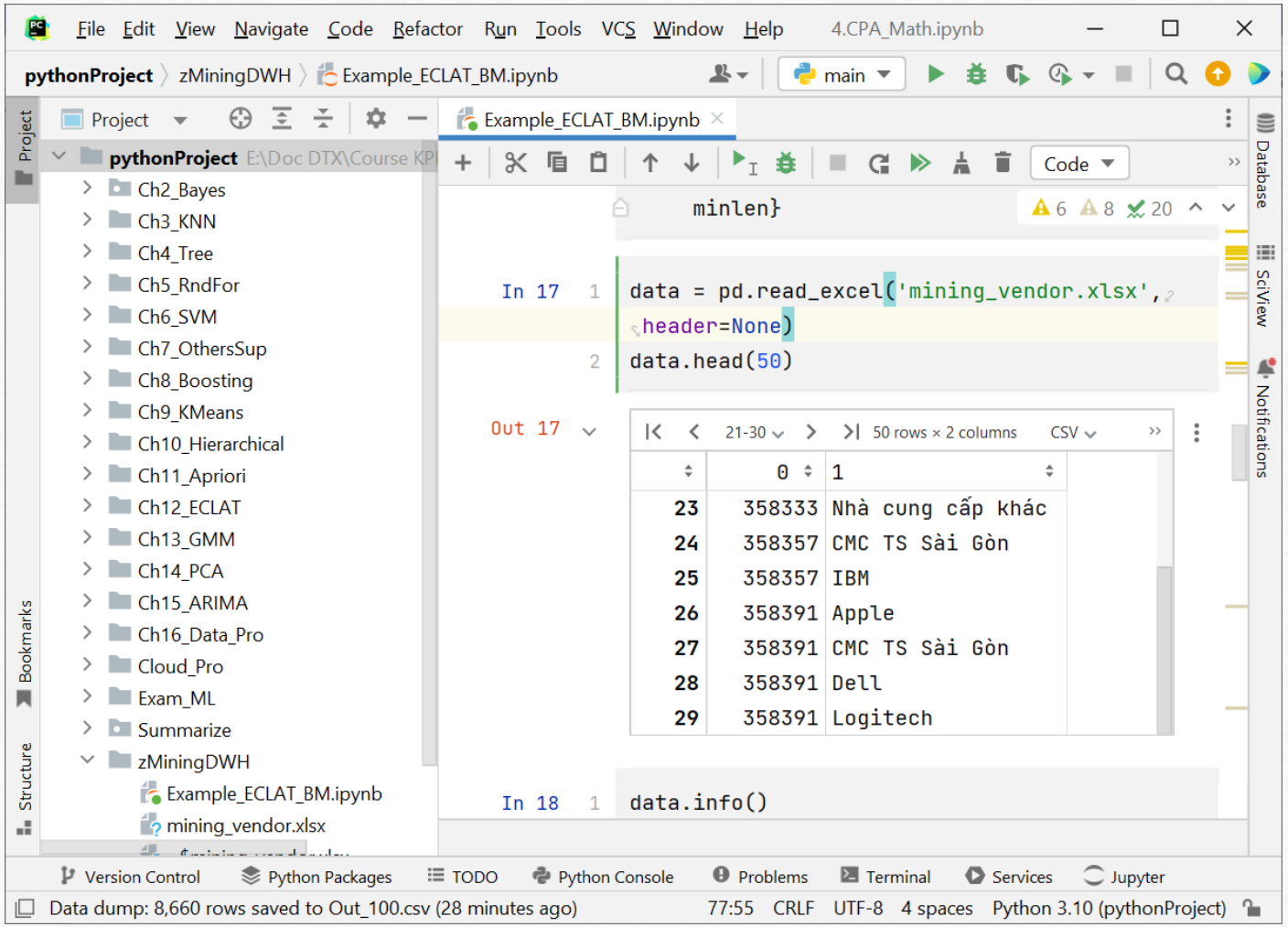
The BI report gets data from DWH and displays a dashboard, which is built following the BSC matrix.

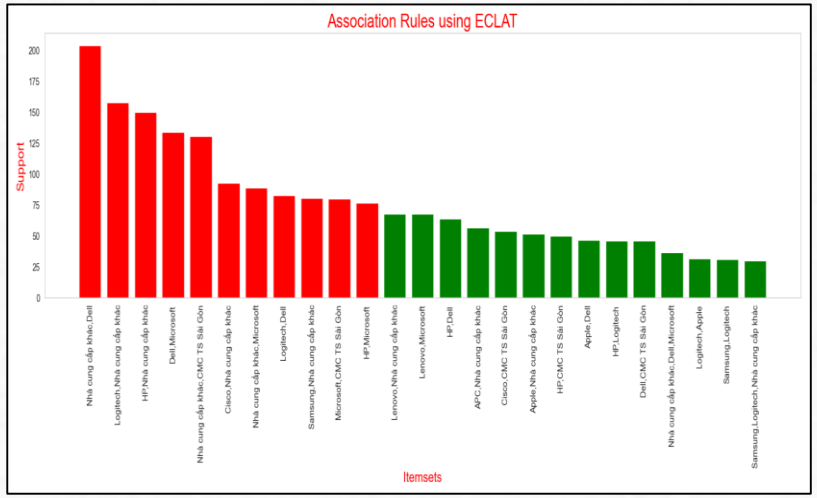




2. Performance BI Project – ML with Python

An example of how to apply ML with Python for diagnostic analysis



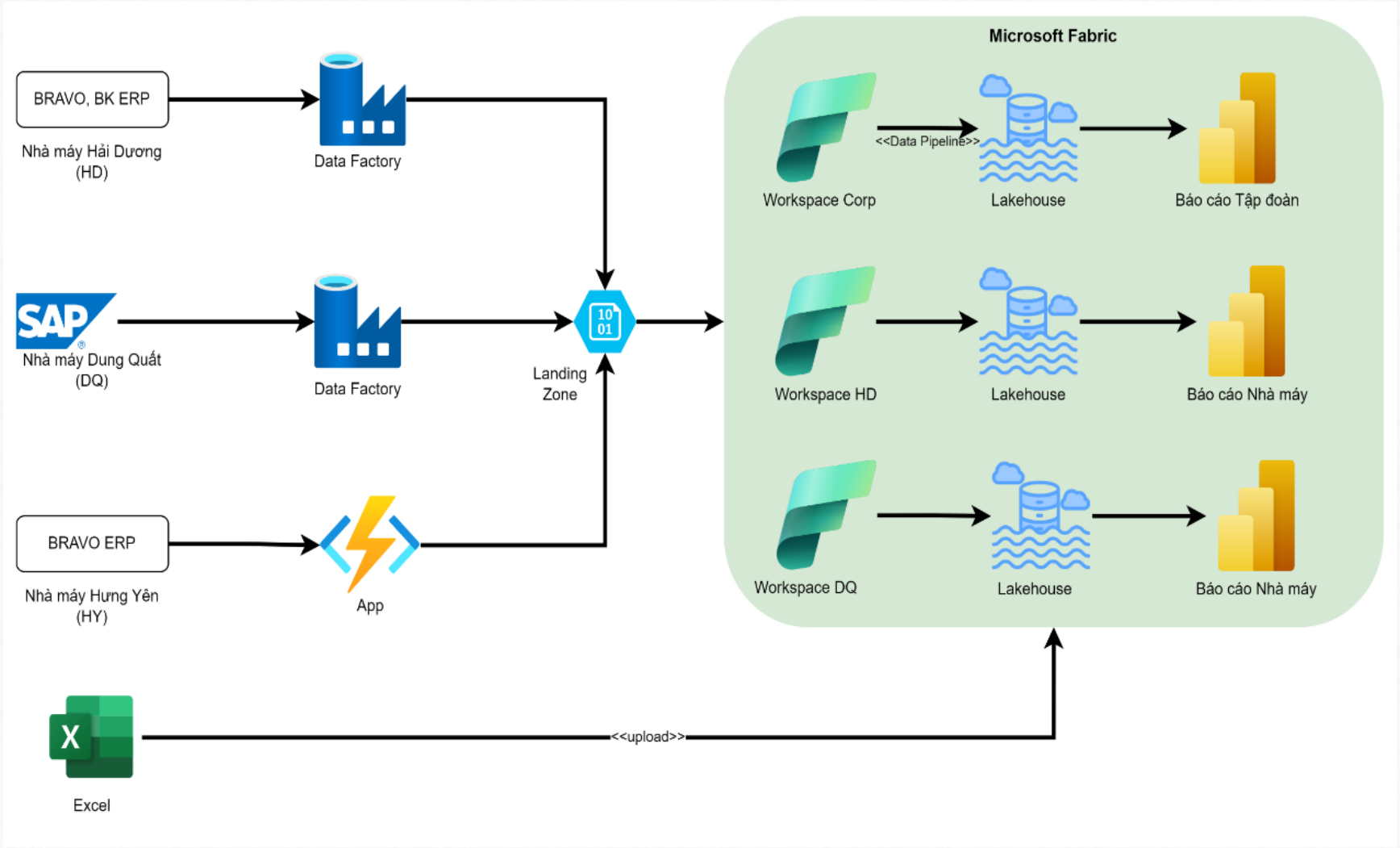



```
# Which items are most frequently bought together
for k, v in result.items():
    if 'CMC TS Sài Gòn' in k:
        print(k, v)
```

```
('Nhà cung cấp khác', 'CMC TS Sài Gòn') 131
('Microsoft', 'CMC TS Sài Gòn') 80
('Cisco', 'CMC TS Sài Gòn') 54
('HP', 'CMC TS Sài Gòn') 50
('Dell', 'CMC TS Sài Gòn') 46
```

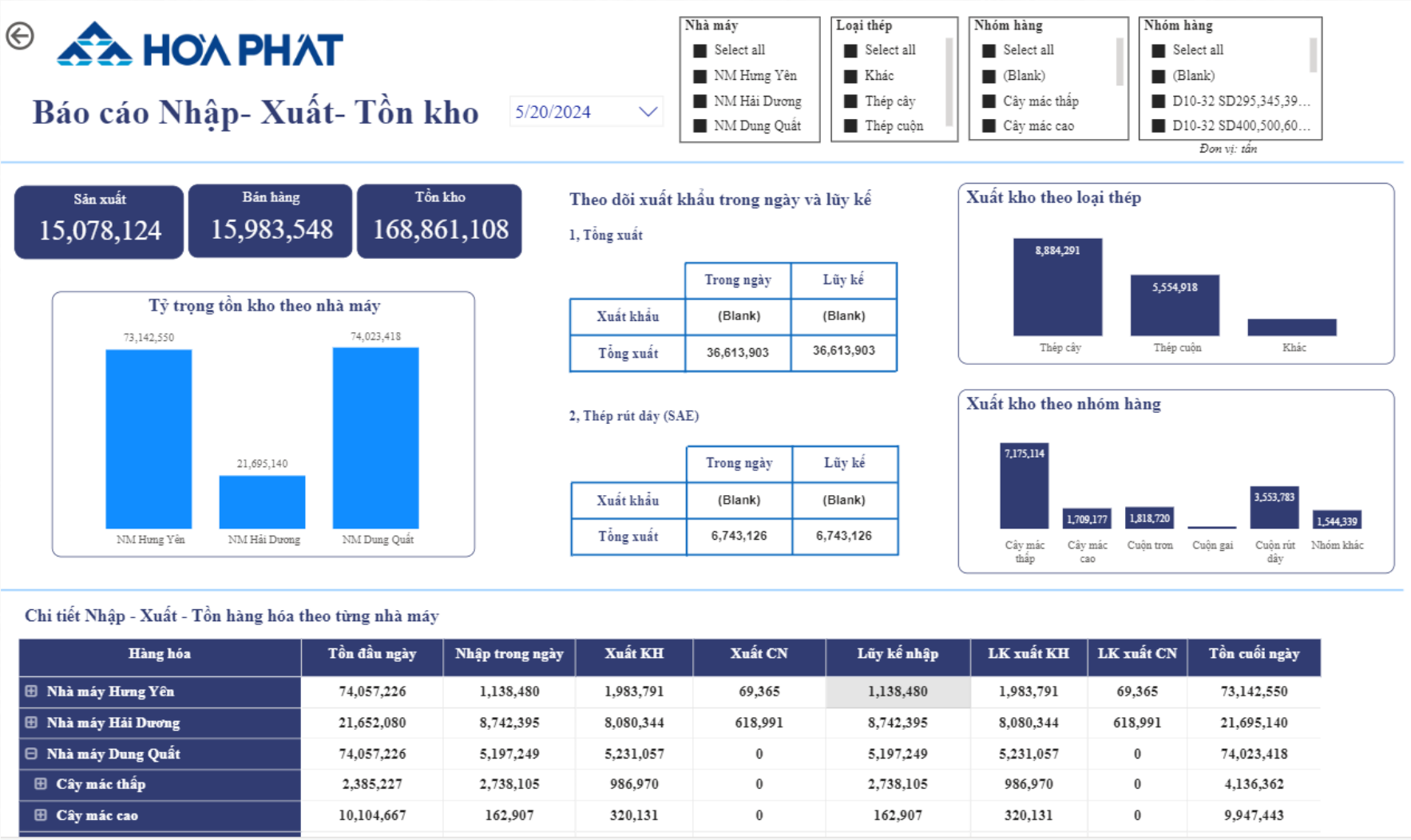
3. Factories BI Project – MS Fabric Platform

ERP and Excel data flow into Microsoft Fabric Lakehouses for corporate and factory reporting.



3. Factories BI Project – Power BI : Dashboard

The dashboard tracks daily inventory movements and stock balances across factories.

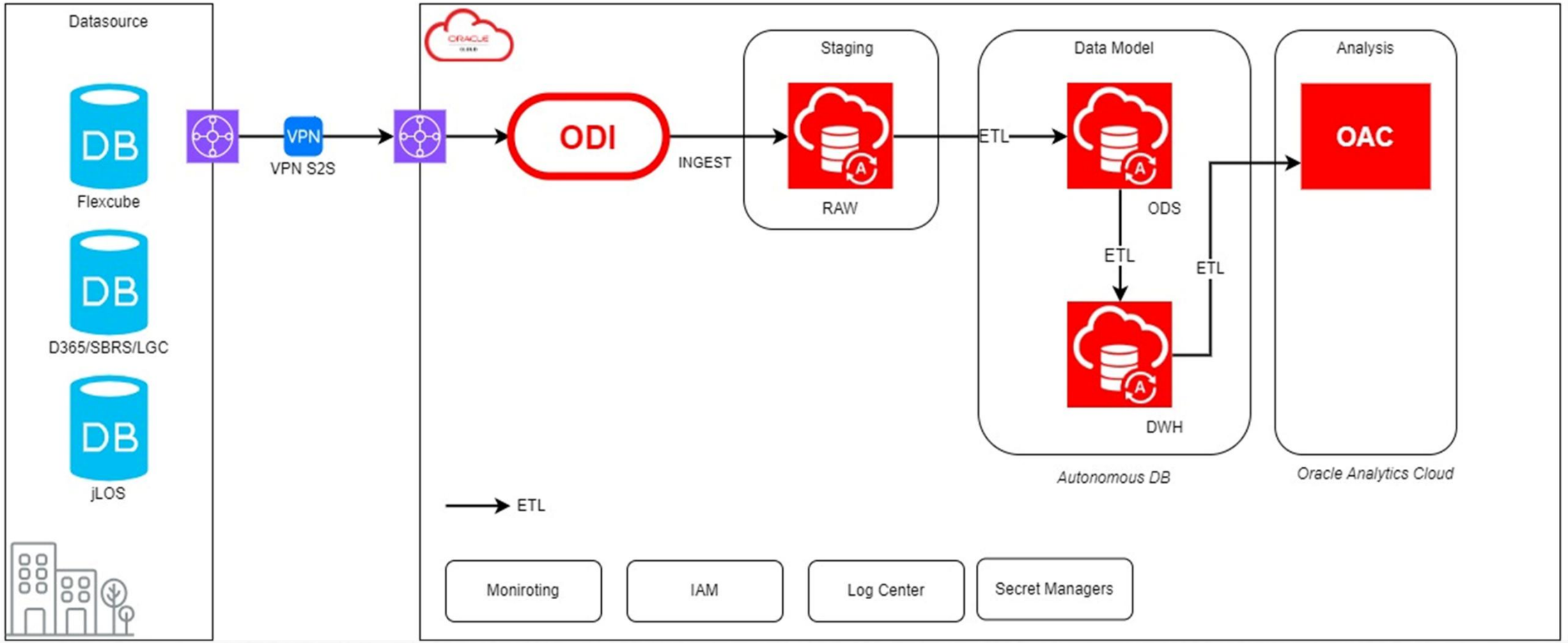


Chi tiết Nhập - Xuất - Tồn hàng hóa theo từng nhà máy

Hàng hóa	Tồn đầu ngày	Nhập trong ngày	Xuất KH	Xuất CN	Lũy kế nhập	LK xuất KH	LK xuất CN	Tồn cuối ngày
⊞ Nhà máy Hưng Yên	74,057,226	1,138,480	1,983,791	69,365	1,138,480	1,983,791	69,365	73,142,550
⊞ Nhà máy Hải Dương	21,652,080	8,742,395	8,080,344	618,991	8,742,395	8,080,344	618,991	21,695,140
⊟ Nhà máy Dung Quất	74,057,226	5,197,249	5,231,057	0	5,197,249	5,231,057	0	74,023,418
⊞ Cây mác thấp	2,385,227	2,738,105	986,970	0	2,738,105	986,970	0	4,136,362
⊞ Cây mác cao	10,104,667	162,907	320,131	0	162,907	320,131	0	9,947,443

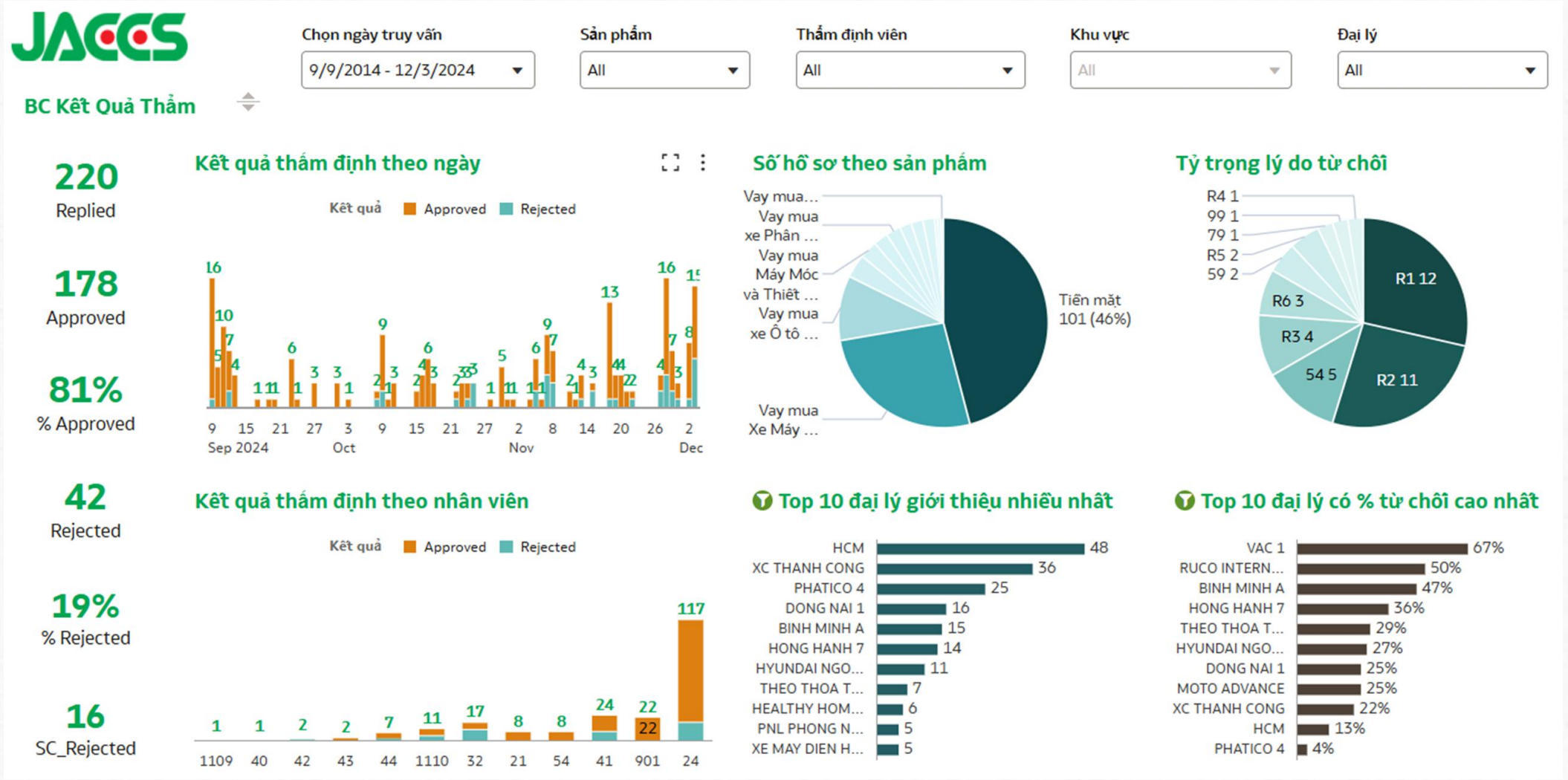
4. Financial Service BI Project – ODI & OAC

ODI ingests financial-service data into Autonomous Database layers that support governed analytics in Oracle Analytics Cloud.



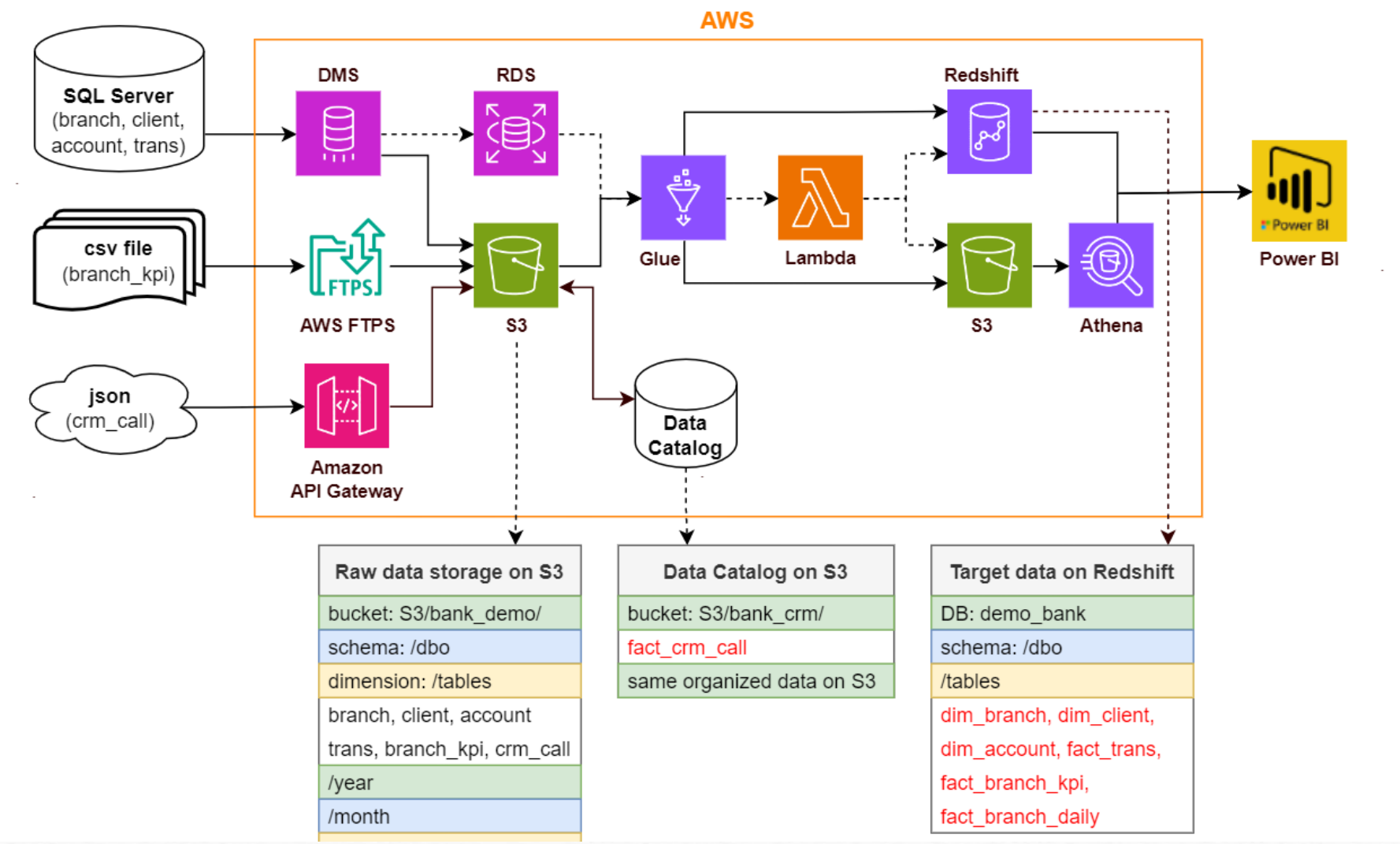
4. Financial Service BI Project – OAC Dashboard

The dashboard tracks appraisal results, approval rates, staff performance, and rejections.



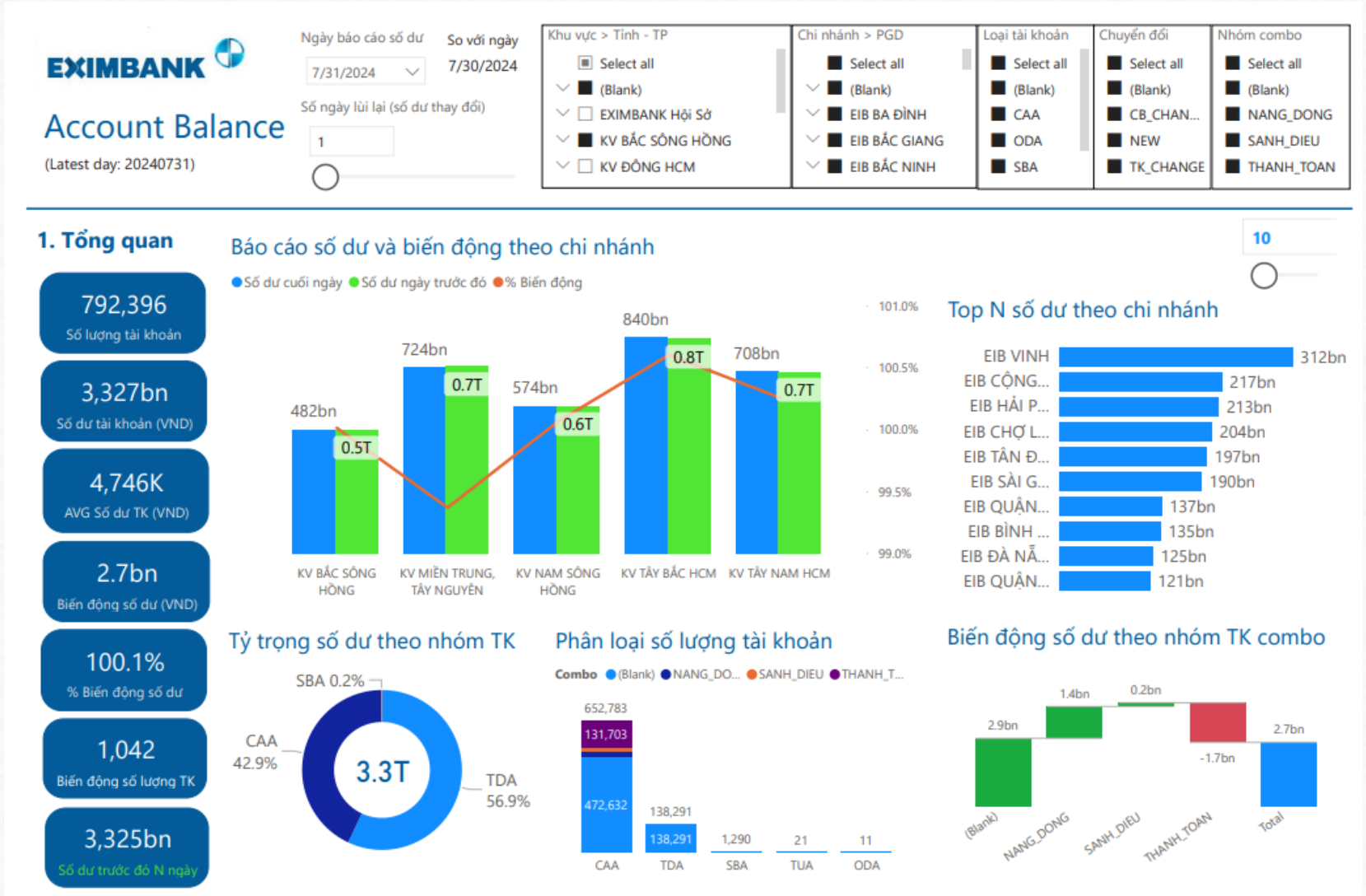
5. Banking BI Project – AWS & Power BI

AWS services ingest transaction, branch KPI, and CRM data into S3 and Redshift for governed Power BI analytics.



5. Banking BI Project – Dashboard Account Balance

The dashboard analyzes account balances by branch, account type, and customer segment.



5. Banking BI Project – Dashboard Transaction Analysis

The dashboard analyzes transaction value and volume by time, branch, currency, and channel.

